E.64 standalone proof review

Statement E.64 (Proved). Full hyperelliptic descent and the exact two CM comparison characters. *Let $\Phi_{n,F}$ be the standard braid automorphism associated to $F=(\lambda,f)\in\widehat{\mathrm{GT}}$. Write*

$$\delta_n=\sigma_1\cdots\sigma_{n-1},\qquad z_n=\delta_n^n,\qquad W_n=\sigma_1\cdots\sigma_{n-2}\sigma_{n-1}^2\sigma_{n-2}\cdots\sigma_1. \tag{E.64.1}$$

For every $g\geq 2$, the full profinite hyperelliptic quotient

$$\widehat{\Upsilon}_g=\widehat{B}_{2g+2}/\overline{\langle\!\langle z_{2g+2},\ W_{2g+2}^2,\ [W_{2g+2},\sigma_i]\rangle\!\rangle} \tag{E.64.2}$$

is preserved by $\Phi_{2g+2,F}$. The induced automorphism fixes its deck involution and induces the standard action on its full genus-zero quotient. In genus two this action has the actual two-boundary sewing comparison

$$\begin{gathered}(\widehat{B}_4\times\widehat{\mathbf{Z}})/\overline{\langle(z_4,-2)\rangle}\ \simeq\ (\widehat{\Gamma}_2)_{\vec{\gamma}},\qquad t^2=z_4,\\ (b,d^a)\longmapsto(\Phi_{4,F}(b),d^{\lambda a}),\qquad t\longmapsto t^{\lambda}.\end{gathered} \tag{E.64.3}$$

Here $\vec{\gamma}$ is the oriented nonseparating cut curve; both boundary twists of the cut surface map to t.

The two CM subgroup maps are equally explicit. In $B_4=\langle a,b,c\rangle$, let

$$\begin{gathered}q:B_4\longrightarrow B_3=\langle a,b\rangle,\qquad q(a)=q(c)=a,\quad q(b)=b,\\ K=\ker q=\langle\xi_1,\xi_2\rangle,\qquad \xi_1=ca^{-1},\quad \xi_2=abac^{-1}b^{-1}a^{-1},\qquad r=ab,\quad s=aba.\end{gathered} \tag{E.64.4}$$

With $B_3^{\mathrm{mor}}=\langle u,v\rangle$, the map

$$\theta(u)=ab,\qquad \theta(v)=b^{-1}cba\quad \textit{identifies}\quad \widehat{B}_3^{\mathrm{mor}}\simeq\widehat{q}^{-1}(\overline{\langle r\rangle}). \tag{E.64.5}$$

It maps $(uv)^3$ to z_4. For the quartic subgroup put

$$\begin{gathered}d=abc,\qquad y=b^{-1}d^2b,\qquad x=(yd)^{-1},\\ G_4=\langle D,Y\mid Y^2=D^4\rangle,\qquad D\mapsto d,\quad Y\mapsto y.\end{gathered} \tag{E.64.6}$$

This identifies $\widehat{G}_4\simeq\widehat{q}^{-1}(\overline{\langle s\rangle})$. The centre maps to $\overline{\langle z_4\rangle}$, and the central quotient is the specified triangle group $C_4\amalg C_2$, with generators (x,y,d) of product one. Both displayed subgroup maps contain exactly the same free Ferrari kernel $\widehat{K}$.

Restrict the remaining assertions to $\lambda=1$. Let A_F,β_F be the unique positive pure frames of E.1 and E.58, lifted to the free complement $\langle a^2,b^2\rangle$ in $\widehat{B}_3$. There is a unique full quartic frame

$$H_F=j(\beta_F)k_F,\qquad k_F\in\widehat{K},\qquad \Phi_{4,F}(d)=H_FdH_F^{-1},\qquad q(H_F)=\beta_F, \tag{E.64.7}$$

where j is the first-three-strand section of q. The normalized actions

$$\begin{aligned}\Psi_{3,F}&=\theta^{-1}\,\mathrm{Inn}(A_F^{-1})\Phi_{4,F}\theta,\\ \Psi_{4,F}&=\mathrm{Inn}(H_F^{-1})\Phi_{4,F}\mid_{\widehat{G}_4}\end{aligned} \tag{E.64.8}$$

are actual group actions. The first fixes u and $(uv)^3$; the second fixes d and z_4.

In the ordered basis (ξ_1,ξ_2) use

$$U=\begin{pmatrix}1&1\\0&1\end{pmatrix},\quad L=\begin{pmatrix}1&0\\-1&1\end{pmatrix},\quad R_+=UL=\begin{pmatrix}0&1\\-1&1\end{pmatrix},\quad S=ULU=\begin{pmatrix}0&1\\-1&0\end{pmatrix}. \tag{E.64.9}$$

Let Q be this literal action of the first $\widehat{B}_3$, and let M_F be the action of $\Phi_{4,F}$ on $\widehat{K}^{\mathrm{ab}}$. Then

$$\begin{gathered} M_F = \begin{pmatrix} 1 & t_F \\ 0 & 1 \end{pmatrix}, \qquad M_F L M_F^{-1} = Q(f)^{-1} L Q(f), \qquad t_F \in \widehat{\mathbf{Z}}, \\ C_3(F) = Q(A_F)^{-1} M_F \in \mathrm{Cent}(R_+), \qquad C_2(F) = Q(\beta_F)^{-1} M_F \in \mathrm{Cent}(S). \end{gathered} \tag{E.64.10}$$

The first covariance characterizes M_F among matrices commuting with U and fixing e_1, over every commutative coefficient ring. In particular the signed rigidity shear with that covariance is literally M_F. Both C_3 and C_2 are continuous characters into the corresponding integral norm-one groups:

	centralizer	*determinant*
R_+	$aI + bR_+$	$a^2 + ab + b^2$
S	$aI + bS$	$a^2 + b^2$.

(E.64.11)

On the E.1 CM matrix fibre their reductions at three are exactly the positive coordinates T, W. The first bit is the sign of T on $(1, -1)$; the second is zero for $W = \pm I$ and one for $W = \pm S$.

Finally let $\Omega_{3,F}$ and $\Omega_{4,F}$ be the original orbifold actions of the same pair on the two displayed triangle groups, lifted with the central powers specified below, and let $B_{3,F}, B_{4,F}$ be their matrices on the same Ferrari basis. The comparisons

$$\mathcal{D}_3(F) = C_3(F) B_{3,F}^{-1}, \qquad \mathcal{D}_4(F) = C_2(F) B_{4,F}^{-1} \tag{E.64.12}$$

are actual continuous norm-one characters. On a Galois element σ with $\chi(\sigma) = 1$, the opened NTY comparison gives the stronger full-group formula

$$\Psi_{3,\sigma} \Omega_{3,\sigma}^{-1} = \mathrm{Inn}(u^{3\rho_K(\sigma)}), \qquad \mathcal{D}_3(\sigma) = (-I)^{\rho_K(\sigma)}. \tag{E.64.13}$$

No value of either comparison character on all of $\widehat{\mathrm{GT}}$ is assigned in this statement.

Proof. We prove the presentation descent and each based subgroup map, then retain the separate orbifold comparisons.

Full hyperelliptic presentation. The exact presentation in [2, Theorems 3.2–3.3, pp.6–7] gives (E.64.2) and its deck involution W_n for $n = 2g + 2$. Its positive generators satisfy exactly our Artin relations; the identity on those generators gives the presentation map from the source's right-composition convention. The needed braid identities are

$$\begin{gathered} z_{\mathrm{first}} = (\sigma_1 \cdots \sigma_{n-2})^{n-1}, \quad z_{\mathrm{last}} = (\sigma_2 \cdots \sigma_{n-1})^{n-1}, \\ z_{\mathrm{last}} = \delta_n z_{\mathrm{first}} \delta_n^{-1}, \qquad W_n = z_n z_{\mathrm{last}}^{-1}. \end{gathered} \tag{E.64.14}$$

The standard action of [5, (1.1)] restricts literally to the first $n - 1$ strands and sends each initial full twist z_k to z_k^{λ}. Thus, for $v_n = \Phi_{n,F}(\delta_n)\delta_n^{-1}$,

$$\Phi_{n,F}(W_n) = v_n W_n^{\lambda} v_n^{-1}. \tag{E.64.15}$$

In the quotient (E.64.2), W_n is a central involution and λ is odd. Hence every stated relator maps to one. Applying the same argument to F^{-1} proves equality of the closed kernels and an actual quotient automorphism. Killing W_n is the usual full sphere mapping-class quotient, so its action is the specified genus-zero action. The completion comparison is exact: the finite quotients of a presented discrete quotient are precisely the finite quotients of its source that annihilate the displayed relators.

The selected genus-two cut and sewing. For the two-boundary torus, the positive three-chain presentation is

$$\Gamma(S_1^2) = B_4 \times \langle d \rangle, \qquad \delta_1 = d, \quad \delta_2 = z_4 d^{-1}. \tag{E.64.16}$$

This is the boundary version of the Birman–Hilden map; [4, Section 6] gives the indicated group, and the three-chain relation identifies the two boundary factors. Gluing the boundary components by an annulus sends both twists to the oriented core twist t. Cutting a homeomorphism that preserves that orientation shows that the kernel is generated by $\delta_1\delta_2^{-1}$: its ambiguity is exactly twisting on the gluing annulus. This gives the discrete version of (E.64.3).

For the intrinsic completed stabilizer, use the actual stabilizer exact sequence in [1, Theorem 2.3] and its genus-at-most-two congruence comparison in Proposition 2.6(i). The completed group on the left of (E.64.3), after killing t, is $\widehat{B}_4/\langle z_4 \rangle = \widehat{P\Gamma}_{1,2}$. Its central t is an injected $\widehat{\mathbf{Z}}$: the map $(b, a) \mapsto \exp(b) + 6a$ annihilates $(z_4, -2)$ and sends t to six. The natural map to the actual stabilizer is the identity on this inertia and on the indicated cut-surface quotient. The two exact sequences prove the claimed isomorphism.

In the genus-two chain $T_1, \dots, T_5$, take $t = T_5$ and $B_4 = \langle T_1, T_2, T_3 \rangle$. The three-chain relation is $(T_1T_2T_3)^4 = T_5^2$. Also $z_5 = (T_1T_2T_3T_4)^5$ is the deck involution: on the sphere it is the twist around five of the six marked points, hence is trivial, and on integral homology it is $-I$. Therefore $f(z_5, T_5^2) = 1$, since its two arguments commute and f is a commutator word. The standard formula gives $\Phi_{6,F}(T_5) = T_5^{\lambda}$, and its restriction to the first three generators is $\Phi_{4,F}$. This proves the actual sewing square, including the two central boundary images.

The Mordell subgroup, before completion. The Ferrari free kernel and its Nielsen action are the literal maps of [6, (2.9), Lemma 4.6]:

	ξ_1	ξ_2
a	ξ_1	$\xi_2\xi_1$
b	$\xi_1\xi_2^{-1}$	ξ_2.

(E.64.17)

The following full braid words retain the central signs in the Mordell triangle:

$$\begin{gathered} \xi_+ = ab, \qquad \xi_{+2} = b^{-1}a^{-1}b^{-1}a^{-1}b^{-1}c^{-1}, \qquad \xi_{+3} = cbab, \\ \theta(uv) = \xi_{+3}, \quad \theta(uvu) = \xi_{+2}^{-1}, \quad \theta((uv)^3) = z_4, \\ \xi_+\xi_{+2}\xi_{+3} = 1, \qquad \xi_{+2}^2 = z_4^{-1}, \quad \xi_{+3}^3 = z_4. \end{gathered} \tag{E.64.18}$$

They prove the Artin relation for θ. Both folded generators equal r. In the source put

$$k_1 = (uv)u^{-2}, \qquad k_2 = u^4(uv)(uv)^{-3}. \tag{E.64.19}$$

Their images are exactly ξ_1, ξ_2. The derived subgroup of B_3 is free of rank two: its image in $C_2 * C_3$ is the index-six free subgroup, and the full centre has nonzero exponent sum. The map from that free group onto K, also free of rank two, is an isomorphism by Hopficity. Thus the two displayed words form a basis, and θ is an isomorphism on the free kernels and infinite cyclic quotients. This proves the discrete version of (E.64.5). In particular the centre map is $z_3^{\mathrm{mor}} \mapsto z_4$, not a discarded central power.

The quartic subgroup, before completion. Direct braid multiplication gives

$$\begin{gathered} y^2 = d^4, \qquad q(d) = s, \quad q(y) = s^2, \quad q(x) = s^{-3}, \\ k_0 = yd^{-2} = \xi_2^{-1}, \qquad k_1 = dyd^{-3} = \xi_1^{-1}, \\ d\xi_1 d^{-1} = \xi_2^{-1}, \qquad d\xi_2 d^{-1} = \xi_1. \end{gathered} \tag{E.64.20}$$

There is a complete presentation verification: define $\alpha(k_0) = k_1$, $\alpha(k_1) = k_0^{-1}$ on the free group. The mutually inverse maps

$$\begin{aligned} G_4 \longrightarrow \langle k_0, k_1\rangle \rtimes_\alpha \mathbf{Z}, \qquad D \mapsto (1,1), \quad Y \mapsto (k_0, 2), \\ k_0 \mapsto YD^{-2}, \qquad k_1 \mapsto DYD^{-3}, \qquad 1 \in \mathbf{Z} \mapsto D \end{aligned} \tag{E.64.21}$$

are homomorphisms because $\alpha^2(k_0) = k_0^{-1}$ and $\alpha^4 = 1$. Equation (E.64.20) identifies both free kernels, and $D \mapsto s$ identifies their cyclic quotients. This proves the discrete quartic assertion. Its centre is $\langle D^4\rangle$, and killing it gives $C_4 * C_2$. The kernel of this free product onto C_4, with $D \mapsto 1$ and $Y \mapsto 2$, is exactly K. Its cusp and centre relation is

$$\begin{aligned} x &= \xi_1 d^{-3}, \\ x^4 z_4^3 &= [\xi_1, \xi_2^{-1}] = \xi_2^{-1}[\xi_1, \xi_2]^{-1}\xi_2. \end{aligned} \tag{E.64.22}$$

The algebraic cover $Y^2 = X^3 - X$, $t = 1/X^2$, has precisely these ramification indices $(4, 2, 4)$, with the origin removed over $t = 0$ and deck generator $(X, Y) \mapsto (-X, iY)$. The displayed triangle generators have deck values $(1, 2, 1)$. The exact braid basis, rather than an unspecified path to that algebraic cover, is used in all matrices below.

Completion of the two actual inclusions. The split extension $B_4 = K \rtimes B_3$ induces the full profinite topology on K. Indeed, every characteristic finite quotient of the finitely generated free group K has a finite automorphism group, so the action factors through a finite quotient of B_3 and supplies the corresponding finite semidirect quotient of B_4. Such characteristic subgroups are cofinal among the finite-index subgroups of K. The section splits the completed quotient, whence $\widehat{B}_4 = \widehat{K} \rtimes \widehat{B}_3$. The closures of $\langle r\rangle$ and $\langle s\rangle$ are each $\widehat{\mathbf{Z}}$, since the exponent-sum maps take them to two and three respectively. The same construction completes $B_3^{\mathrm{mor}} = K \rtimes \mathbf{Z}$ and (E.64.21). It proves both stated completed subgroup isomorphisms.

Full accessory frames and their cocycles. The E.1 pure frames give the equalities modulo the first three-strand centre. Their exponent sums agree upstairs, and that centre has exponent sum six. Hence the exact lifted equalities are

$$\Phi_{3,F}(r) = A_F r A_F^{-1}, \qquad \Phi_{3,F}(s) = \beta_F s \beta_F^{-1}. \tag{E.64.23}$$

The Ferrari fold and section satisfy $q\Phi_{4,F} = \Phi_{3,F}q$ and $\Phi_{4,F}j = j\Phi_{3,F}$ on their literal generators. Thus normalization by A_F gives the first map in (E.64.8).

For the quartic map, E.39 supplies a full conjugator H for d. The centralizer of s in $\widehat{B}_3$ is $\overline{\langle s\rangle}$: after quotienting by $z_3 = s^2$, the factor-centralizer theorem [3, Theorem B', Section 4, p.159] gives $C_{C_2 \amalg C_3}(\bar{s}) = \langle \bar{s}\rangle$. Consequently $q(H) = \beta_F s^a$ for a unique $a \in \widehat{\mathbf{Z}}$. Replace H by Hd^{-a}. This preserves its conjugacy equation and gives (E.64.7).

This frame is unique. The same theorem gives $C_{C_4 \amalg C_2}(\bar{d}) = \langle \bar{d}\rangle$. A member of $\widehat{K}$ centralizing d therefore has trivial image in the central quotient, since $\widehat{K} \cap \langle \bar{d}\rangle = 1$ there. It is then in $\overline{\langle d^4\rangle} \cap \widehat{K} = 1$, as $q(d^4) = s^4$ has infinite profinite order. Thus $C_{\widehat{K}}(d) = 1$.

The frames are continuous by the compact conjugation maps and their uniqueness. That uniqueness gives the exact identities

$$\begin{aligned} A_{FG} = \Phi_{3,F}(A_G)A_F, \qquad \beta_{FG} = \Phi_{3,F}(\beta_G)\beta_F, \\ H_{FG} = \Phi_{4,F}(H_G)H_F. \end{aligned} \tag{E.64.24}$$

For the last one, the right side has precisely the required projection β_{FG}. Equations (E.64.24) prove that both normalized maps in (E.64.8) are group actions. They fix the stated generators and centres literally.

Integral homology and the rigidity identification. Equation (E.64.17) gives $Q(a) = U$, $Q(b) = L$. In the standard formula, $e = f(z_3, c^2)$ lies in $\widehat{K}$, because $q(e) = f(z_3, a^2) = 1$. Both its arguments commute with a, so e does also. Therefore $\Phi_{4,F}(\xi_1) = e^{-1}\xi_1 e$, and the first column of M_F is e_1. Since $\Phi_{4,F}(a) = a$, covariance gives $M_F U = U M_F$. This proves the asserted unipotent shear. Covariance with $\Phi_{3,F}(b) = f(a^2, b^2)^{-1} b f(a^2, b^2)$ gives the first identity in (E.64.10).

Here is the promised comparison over every coefficient ring. For $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$,

$$AU - UA = \begin{pmatrix} -c & a-d \\ 0 & c \end{pmatrix}, \qquad AL - LA = \begin{pmatrix} -b & 0 \\ a-d & b \end{pmatrix}. \tag{E.64.25}$$

Thus the joint centralizer consists of scalars, over any commutative ring. The quotient of two matrices satisfying the same covariance commutes with U, L; fixing e_1 makes that scalar one. This proves the exact equality with the signed rigidity shear, including over $\mathbf{Z}_\ell$ and $\widehat{\mathbf{Z}}$.

For the finite E.61 convention put $t = 8\rho$ and

$$F_\rho = \begin{pmatrix} 1 & -t \\ -t & 1 + t^2 \end{pmatrix}, \qquad V = U^t, \qquad R_0 = -R_+. \tag{E.64.26}$$

Direct integer multiplication gives

$$VLV^{-1} = F_\rho^{-1} L F_\rho, \qquad VSV^{-1} = SF_\rho, \qquad VR_0V^{-1} = F_\rho^{-1} R_0. \tag{E.64.27}$$

Consequently $t_F = 8\rho_F \pmod{12}$, with precisely this orientation. On the E.1 matrix fibre $F_\rho = I$, so $M_F = I \pmod 3$.

The first normalized action fixes u, whose matrix on the free kernel is R_+, and the second fixes d, whose matrix is S. Their homology matrices are exactly C_3, C_2: the factor k_F of (E.64.7) is an inner automorphism of $\widehat{K}$ and hence has trivial abelianization action. The frame cocycles and covariance prove multiplicativity. Solving the four entry equations gives (E.64.11) over every commutative ring; the determinants are one. At three the norm-one groups have the six and four elements used in E.1. Their matrices are the positive T, W by (E.64.23) and $M_F = I \pmod 3$, with the exact two bit maps stated above.

The original orbifold maps, with their central lifts. In the Mordell source put $x = u$, $y = (uvu)^{-1}$, $z = uv$. Then $xyz = 1$ and $y^2 = z_3^{-1}$, $z^3 = z_3$. In the quartic source use the three elements $(x, y, z) = (x, y, d)$ of (E.64.6); here $y^2 = z^4 = z_4$. In either source the original puncture-loop action has the same actual formula

$$\begin{aligned} \Omega_F(x) &= x^\lambda, \\ \Omega_F(y) &= f(y, x) y^\lambda f(x, y), \\ \Omega_F(z) &= x^m f(z, x) z^\lambda f(x, z) x^{-m}, \qquad m = (\lambda - 1)/2. \end{aligned} \tag{E.64.28}$$

Modulo the stated centre, its product relation is the gentle equation, and the finite-order generators retain their orders. The product discrepancy upstairs is central. In the Mordell case the exponent sum sends that centre to six; in the quartic case the map $(z, y, x) \mapsto (1, 2, -3)$ sends it to four. The discrepancy has value zero, hence is one. The central power relations map to their λ powers by conjugation, so (E.64.28) gives actual lifts. The inverse pair and the same central detection prove invertibility and the composition law. This proves the orbifold maps as separate homomorphisms; it makes no unstated identification with the standard three-strand braid action.

At multiplier one, the Mordell map fixes $x = u$ and the quartic image of d is conjugate to d by $f(d, x) \in K$. Their matrices therefore commute respectively with R_+ and S. Their determinants are one. For example the exact cusp relations are

$$[k_1, k_2]u^6 = z_3^{\mathrm{mor}}, \qquad x^4 z_4^3 = [\xi_1, \xi_2^{-1}], \tag{E.64.29}$$

so both maps fix a primitive commutator in the second lower-central quotient. Their homology representations $B_{3,F}, B_{4,F}$ have the same bases as C_3, C_2. All four matrices lie in the respective commutative centralizers. Their quotients are therefore the actual characters (E.64.12). For $B_{3,F}$, this is exactly the triangle Tate action of [6, Proposition 4.4].

The evaluated Galois comparison and its scope. The specified path in [6, Proposition 4.11, Section 4.12, Proposition 4.13], at multiplier one, has

$$u_\sigma = h_\sigma(a^2, b^2)^{-1}(ab)^{3\rho_K(\sigma)} = A_\sigma z_3^{\rho_K(\sigma)}. \tag{E.64.30}$$

Here $h_\sigma = A_\sigma^{-1}$ is the equianharmonic frame; the harmonic frame denoted g_σ in that source is β_σ^{-1}. The source's full braid-generator equations give $\Phi_{4,\sigma}\theta = \mathrm{Inn}(u_\sigma)\theta\Omega_{3,\sigma}$. Since $z_3 = (ab)^3 = \theta(u^3)$, normalizing by A_σ gives (E.64.13) at the full group level. On homology the first-three-strand centre acts by $R_+^3 = -I$. The total centre instead acts trivially, as is also recorded by the exact boundary word

$$[\xi_1, \xi_2] = z_4 z_3^{-2}. \tag{E.64.31}$$

These explicit centre maps account for the sign in the Galois comparison. Neither (E.64.15) nor the selected sewing square replaces the unevaluated full-GT values of (E.64.12) by (E.64.30).

Retained source audit and exact certificates. The source audit preserves the actual capping map from $B_{2g+1}/\langle z_{2g+1}^2 \rangle$ as a branch-point stabilizer of index $2g+2$. It also preserves the identity and $\mathrm{Inn}(\tau_{\gamma_i})$ as two different based automorphisms agreeing on the oriented curve stabilizer in the setting of Boggi v6 Lemma 6.16. The present proof uses the full presentation and the explicit sewing maps above; it does not require either printed identification. The exact audit and the tested rejected double-centre alternative remain in the research files.

There are two finite replay programs, sharing prefix `bc_`: `hyperelliptic_sewing.py` and `cm_norm_characters.py`. The former verifies 45 literal braid identities with the faithful Artin action and the separate integral genus-two matrix comparisons. The latter verifies 14 quartic braid identities, the independent free-by-cyclic presentation, the signed covariances, and both positive CM tables. Its checks of centralizers over $\mathbf{Z}/m$, $2 \le m \le 12$, supplement the coefficient-ring proof (E.64.25). Each program retains its source hashes and complete `.json` and `.out` records. They assign no full-GT value to the comparison characters beyond the maps proved here. □

References

[1] Marco Boggi. Automorphisms of profinite mapping class groups, 2026. verified: complete primary v6 dated 22 April 2026 opened and saved; E.64 uses only the stabilizer exact sequence in Theorem 2.3 and the genus-at-most-two congruence comparison in Proposition 2.6(i). The separate capping-image and based-uniqueness issues in Proposition 6.9 and Lemma 6.16 are preserved with exact witnesses in the research audit and are not used as proof steps.

[2] Hisaaki Endo and Seiichi Kamada. Counting dirac braid relators and hyperelliptic lefschetz fibrations, 2016. verified: primary v2 dated 5 October 2016 opened, Theorems 3.2–3.3 on printed pp.6–7 give the full sphere and hyperelliptic presentations and the exact central involution; identity on the displayed positive generators supplies the convention map used in E.64.

[3] Wolfgang Herfort and Luis Ribes. Torsion elements and centralizers in free products of profinite groups. *Journal für die reine und angewandte Mathematik*, 358:155–161, 1985. verified: complete primary article opened independently by root and both proof readers. Theorem B' in Section 4, printed p. 159, and its proof give the factor-centralizer equality used in E.58; the precise applications are to the actual factors C2 and C3.

[4] Dan Margalit and Rebecca R. Winarski. The birman–hilden theory. verified: complete author PDF opened; Section 4 gives the full genus-two central quotient and presentation, Section 6 gives the boundary-torus braid comparison used with its exact three-chain relation in E.64.

[5] Hiroaki Nakamura and Leila Schneps. On a subgroup of the grothendieck–teichmüller group acting on the tower of profinite teichmüller modular groups. *Inventiones Mathematicae*, 141(3):503–560, 2000. verified: independently opened author-uploaded complete text; section 5 Definition 5.1 and Proposition 5.3, printed pp.522–524; Remark 6.2, printed pp.525–526; the latter comparison explicitly retains relation (IV). The author-hosted version was also opened at Section 1, formula (1.1) and its quotient-action paragraph, and Theorem 2.5 on manuscript page 11; the latter matrix formula is used only on Galois elements. The arithmetic sign was checked again at the positive-root definition at the beginning of Section 2, the generator-convention paragraph after (2.1), (2.4), and Lemma 2.3 on author-PDF p.9; the exact double-cover and inversion maps are proved in D.5.

[6] Hiroaki Nakamura, Hiroshi Tsunogai, and Seidai Yasuda. Harmonic and equianharmonic equations in the grothendieck–teichmueller group. iii. *Journal of the Institute of Mathematics of Jussieu*, 9(2):431–448, 2010. verified: opened the complete author-uploaded primary PDF; Proposition 2.19, Proposition 4.4, Proposition 4.13, Theorem 4.16, and the scope statement on printed p.432. Proposition 4.4 is stated for GT; Theorem 4.16 is stated for Galois elements.